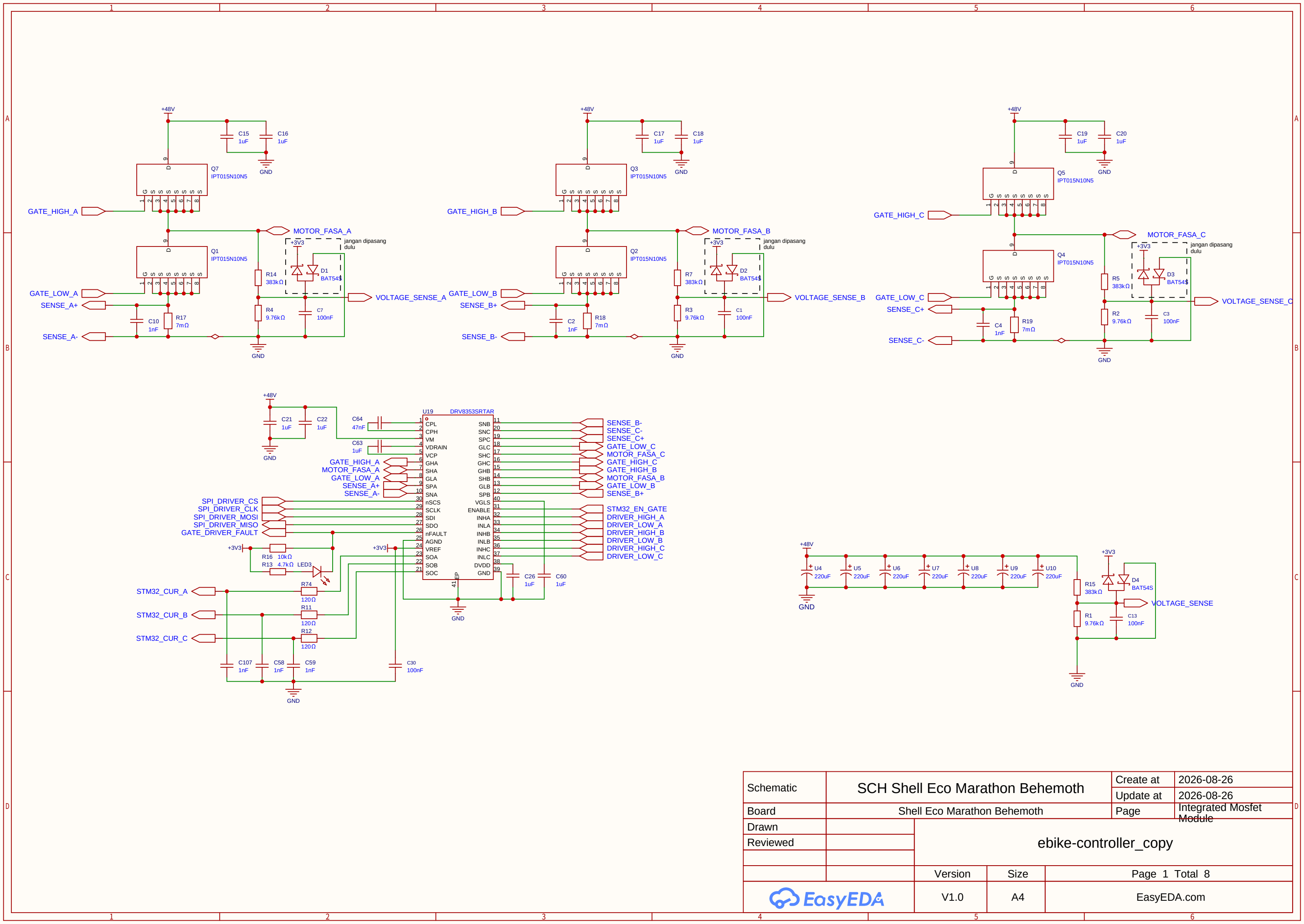
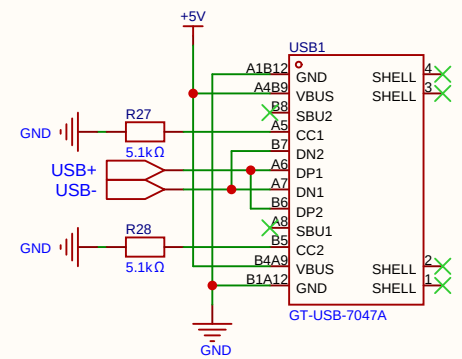
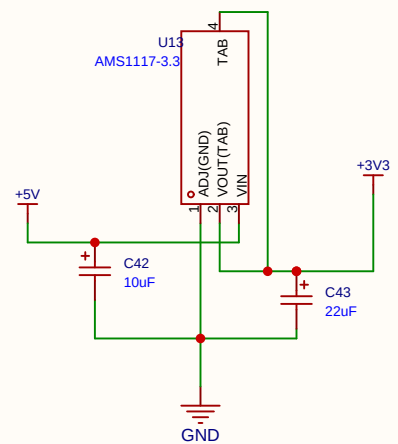
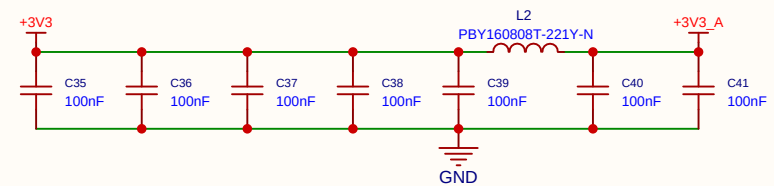
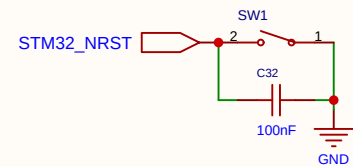
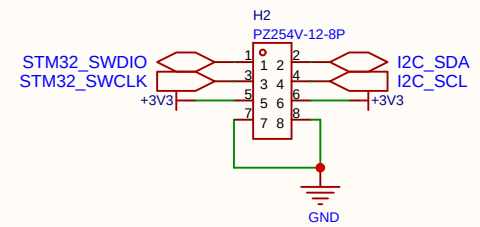
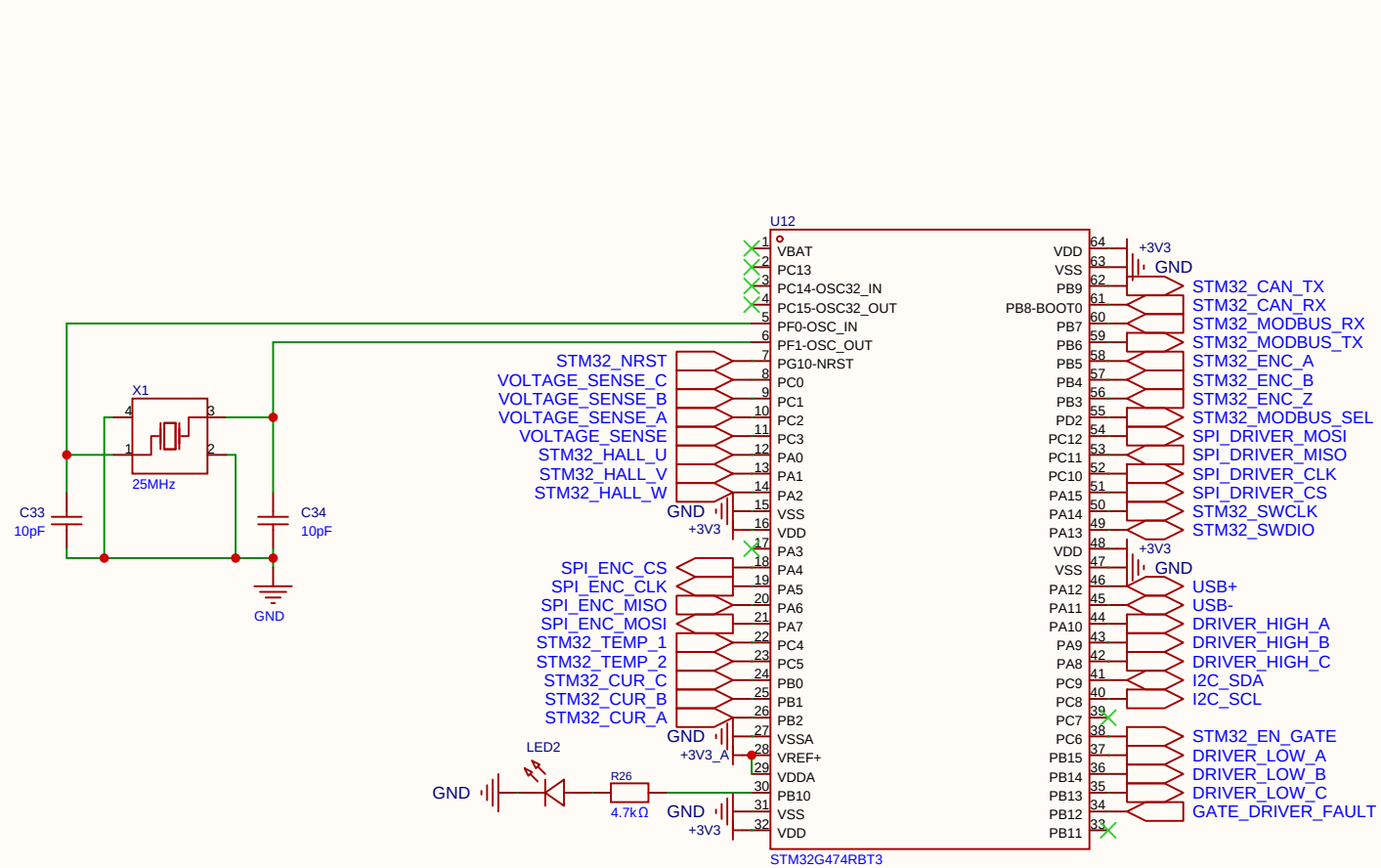




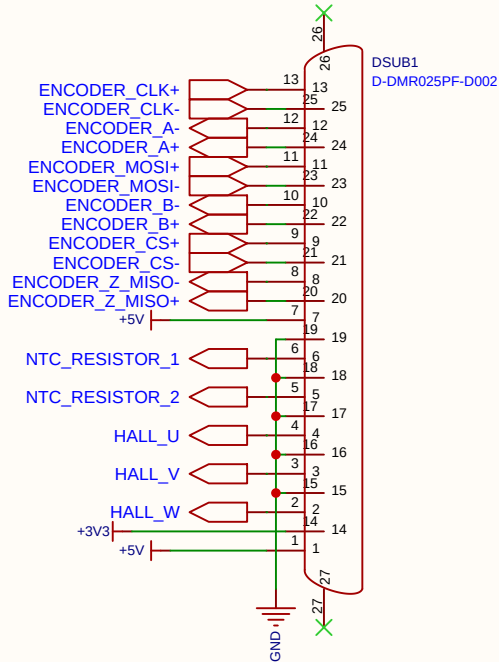
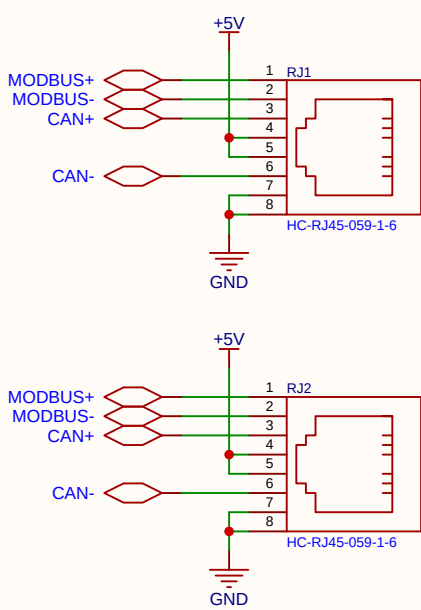
Schematic	SCH Shell Eco Marathon Behemoth	Create at	2026-08-26	
		Update at	2026-08-26	
Board	Shell Eco Marathon Behemoth		Page	Integrated Mosfet Module
Drawn	ebike-controller_copy			
Reviewed				
		Version	Size	Page 1 Total 8
		V1.0	A4	EasyEDA.com



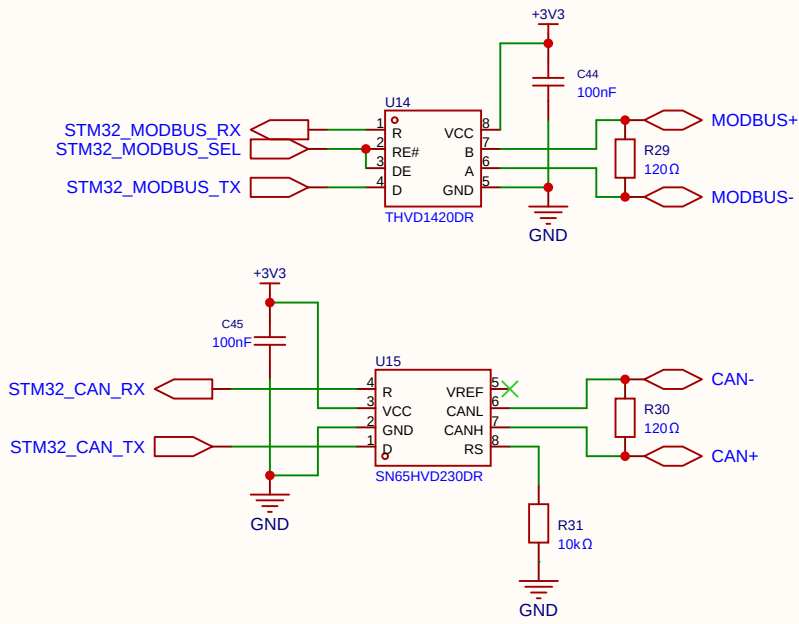
Schematic	SCH Shell Eco Marathon Behemoth	Create at	2026-08-26	
		Update at	2026-08-26	
Board	Shell Eco Marathon Behemoth		Page	STM32 min requirements
Drawn	ebike-controller_copy			
Reviewed				
		Version	Size	Page 2 Total 8
		V1.0	A4	EasyEDA.com

Pinouts [edit]

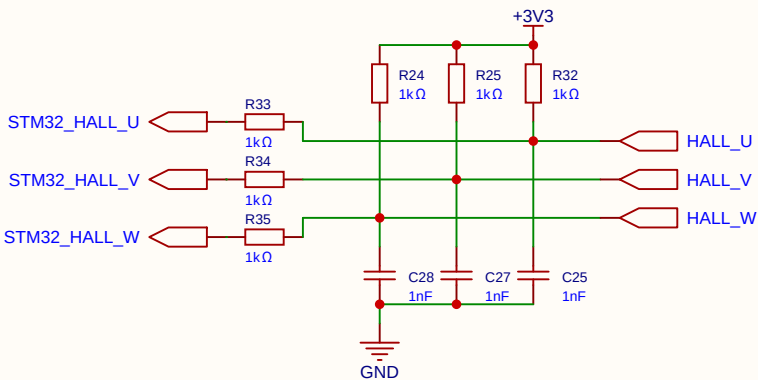
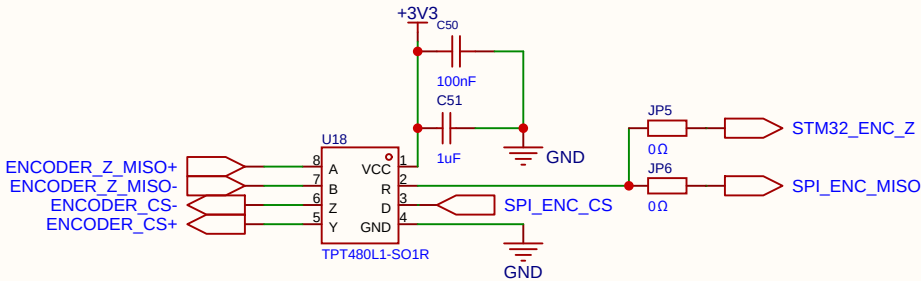
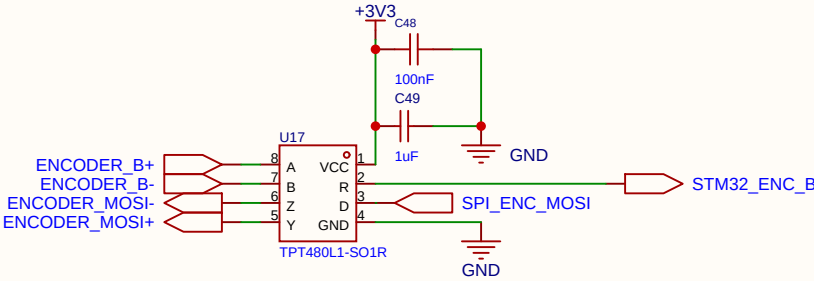
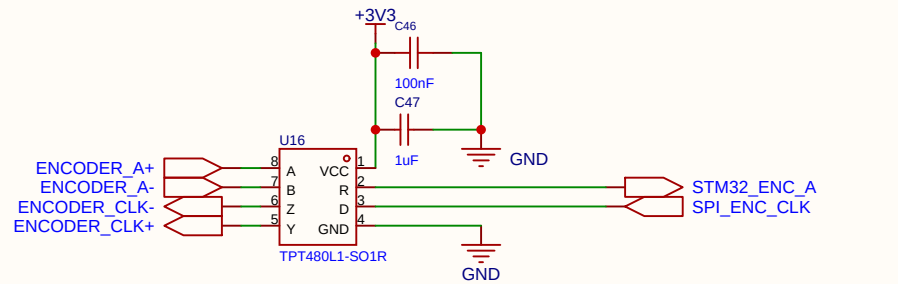
802.3af/at/bt pin assignments from the Power Sourcing Equipment (PSE) perspective (MDI-X)												
Pins at switch	T568A color	T568B color	10/100 mode B, DC on spares		10/100 mode A, DC & data		1000 mode B, DC & bi-data		1000 mode A, DC & bi-data		1000 mode A+B (4PPoE), DC & bi-data ^[note 1]	
Pin 1	 green stripe	 orange stripe	Rx +		Rx +	DC +	TxRx A +		TxRx A +	DC +	TxRx A +	DC +
Pin 2	 solid green	 solid orange	Rx −		Rx −	DC +	TxRx A −		TxRx A −	DC +	TxRx A −	DC +
Pin 3	 orange stripe	 green stripe	Tx +		Tx +	DC −	TxRx B +		TxRx B +	DC −	TxRx B +	DC −
Pin 4	 solid blue			DC +	Unused		TxRx C +	DC +	TxRx C +		TxRx C +	DC +
Pin 5	 blue stripe			DC +			TxRx C −	DC +	TxRx C −		TxRx C −	DC +
Pin 6	 solid orange	 solid green	Tx −		Tx −	DC −	TxRx B −		TxRx B −	DC −	TxRx B −	DC −
Pin 7	 brown stripe			DC −	Unused		TxRx D +	DC −	TxRx D +		TxRx D +	DC −
Pin 8	 solid brown			DC −			TxRx D −	DC −	TxRx D −		TxRx D −	DC −
Notes:												
1. ^ Only supported by 802.3bt for devices that identify as the newly added Type 3 or Type 4. ^[63]												



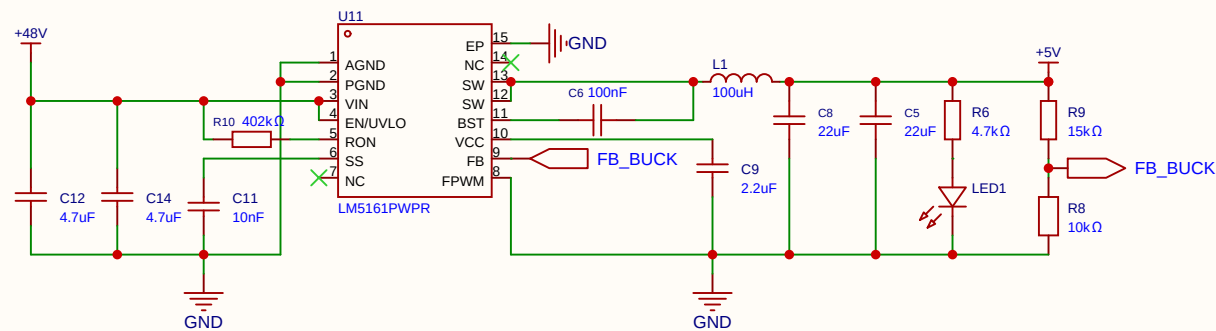
Schematic	SCH Shell Eco Marathon Behemoth	Create at	2026-08-25	
		Update at	2026-08-26	
Board	Shell Eco Marathon Behemoth		Page	Konektor
Drawn	ebike-controller_copy			
Reviewed				
		Version	Size	Page 3 Total 8
		V1.0	A4	EasyEDA.com



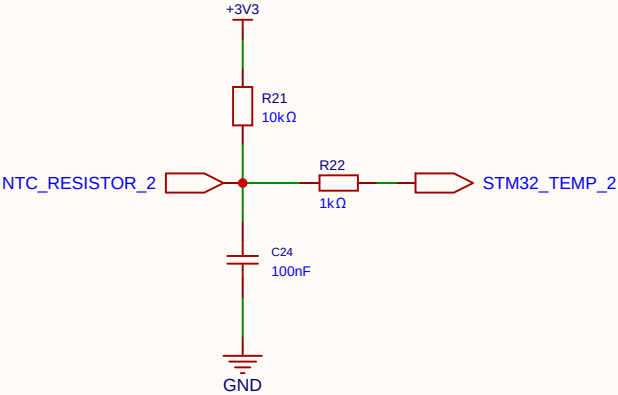
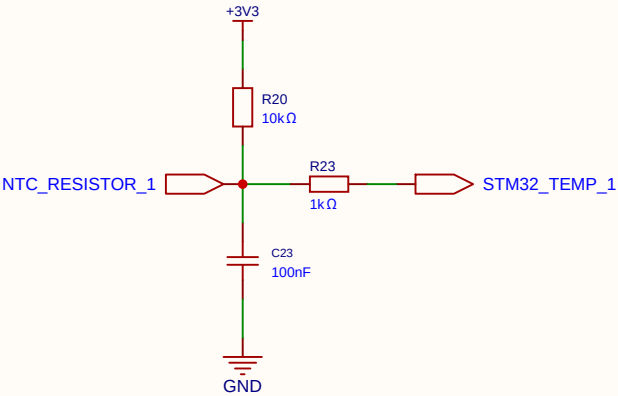
Schematic	SCH Shell Eco Marathon Behemoth			Create at	2026-08-25
				Update at	2026-08-25
Board	Shell Eco Marathon Behemoth			Page	Interface
Drawn		ebike-controller_copy			
Reviewed					
		Version	Size	Page 4 Total 8	
		V1.0	A4	EasyEDA.com	



Schematic	SCH Shell Eco Marathon Behemoth	Create at	2026-08-25	
		Update at	2026-08-26	
Board	Shell Eco Marathon Behemoth		Page	Encoder interface
Drawn	ebike-controller_copy			
Reviewed				
		Version	Size	Page 5 Total 8
		V1.0	A4	EasyEDA.com



Schematic	SCH Shell Eco Marathon Behemoth	Create at	2026-08-24	
		Update at	2026-08-24	
Board	Shell Eco Marathon Behemoth		Page	Power
Drawn		ebike-controller_copy		
Reviewed				
		Version	Size	Page 6 Total 8
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	SCH Shell Eco Marathon Behemoth			Create at	2026-08-25
				Update at	2026-08-26
Board	Shell Eco Marathon Behemoth			Page	Temp Sensor
Drawn		ebike-controller_copy			
Reviewed					
		Version	Size	Page 7 Total 8	
		V1.0	A4	EasyEDA.com	

General Specifications
Microcontroller: STMicroelectronics STM32G474 (Math Accelerator, HRTIM, Triple-ADC)
Gate Driver: Texas Instruments DRV8353SR (100V, Smart Gate Drive, SPI)
Power Stage: Infineon IPT015N10N5 (100V, 1.5 mΩ , TO-Leadless (TOLL))
Logic Power Supply: Texas Instruments LM5161PWPR (100V Input, Fly-Buck Topology)
Control Algorithm: Field Oriented Control (FOC) dengan Regenerative Braking & Coast-and-Burn capability.
Electrical Ratings
Nominal DC Bus Voltage: 48V - 60V DC (Optimal for SEM)
Absolute Maximum Voltage: 100V DC (Hardware destruction limit)
Overvoltage Protection (OVP) Trigger: Disarankan di-set pada 80V - 85V (melalui pembacaan VBUS di ADC STM32) untuk memutus PWM regen.
Continuous Phase Current: 50A RMS
Peak Phase Current: 120A (selama < 5 detik)
Switching Frequency: 20 kHz - 40 kHz (Dapat disesuaikan. Semakin tinggi frekuensi, switching loss naik namun current ripple turun).

Schematic	SCH Shell Eco Marathon Behemoth	Create at	2026-08-24	
		Update at	2026-08-24	
Board	Shell Eco Marathon Behemoth		Page	Dokumentasi
Drawn		ebike-controller_copy		
Reviewed				
		Version	Size	Page 8 Total 8
		V1.0	A4	EasyEDA.com